A One-Scalar Implicit Pac-Man Function

An exact level-one equation and reproducible Python implementation

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Abstract

A Pac-Man outline can be specified directly by the single implicit equation $\max(x^2 + y^2, 1 + (\tan\alpha)x - |y|) = 1$. This note derives the equation, proves that its solution set is precisely a circular arc joined to two radial mouth edges, and gives the corresponding filled region, geometric properties, and runnable Python code. The level-one form is an exact algebraic reformulation of the earlier level-zero field, not an approximation or a new geometry. Related published constructions are discussed without claiming mathematical priority.

1 A single equation for the complete outline

For a mouth half-angle $0 < \alpha < \pi/2$, define

$$\boxed{G_\alpha(x, y) = \max\big(x^2 + y^2,\ 1 + (\tan\alpha)x - |y|\big)\,.} \tag{1}$$

The complete Pac-Man outline is the planar solution set of

$$\boxed{\max\big(x^2 + y^2,\ 1 + (\tan\alpha)x - |y|\big) = 1.}$$

The most familiar special case has $\alpha = \pi/4$, giving a 90° mouth:

$$\boxed{\max(x^2 + y^2,\ 1 + x - |y|) = 1.}$$

The filled region and its boundary are, respectively,

$$P_\alpha = \{(x, y) : G_\alpha(x, y) \le 1\}, \qquad \partial P_\alpha = \{(x, y) : G_\alpha(x, y) = 1\}.$$

Points in the interior satisfy $G_\alpha < 1$; exterior points, including the open mouth wedge, satisfy $G_\alpha > 1$.

What kind of function is this? For fixed α, $G_\alpha : \mathbb{R}^2 \to \mathbb{R}$ is a genuine scalar-valued function: each input pair has exactly one output. The equation $G_\alpha = 1$ defines an implicit curve, in the same sense as a heart-shaped implicit equation. It is not a graph $y = f(x)$. The full graph $z = G_\alpha(x, y)$ is a surface; intersecting it with $z = 1$ recovers the same planar outline. Writing the curve directly as an equation does not remove this level-set interpretation.

2 Why the equation works

For any real numbers A, B, $\max(A, B) = 1$ means that $A \le 1$, $B \le 1$, and at least one of them equals 1. Also,

$$G_\alpha \le 1 \iff x^2 + y^2 \le 1 \text{ and } |y| \ge (\tan\alpha)x, \tag{2}$$

$$P_\alpha = \{x^2 + y^2 \le 1\} \setminus \{x > 0,\ |y| < (\tan\alpha)x\}. \tag{3}$$

The second constraint removes a right-facing wedge of full angle 2α. On the left half-plane it holds automatically. Tangent occurs because the slope of a ray making angle α with the positive horizontal axis is $y/x = \tan\alpha$; the absolute value supplies the lower ray as well.

There are exactly two types of points satisfying $G_\alpha = 1$:

1. **Circular arc:** $x^2+y^2 = 1$ and $|y| \geq (\tan\alpha)x$. Equivalently, $(x, y) = (\cos\theta, \sin\theta)$ for $\alpha \leq \theta \leq 2\pi-\alpha$.
2. **Mouth edges:** $|y| = (\tan\alpha)x$ and $x^2 + y^2 \leq 1$. Since $\tan\alpha > 0$, this forces $x \geq 0$, and hence $y = \pm(\tan\alpha)x$ with $0 \leq x \leq \cos\alpha$.

These join at the lips $(\cos\alpha, \pm\sin\alpha)$, and the mouth edges meet at the origin. There are no extra branches or isolated solutions.

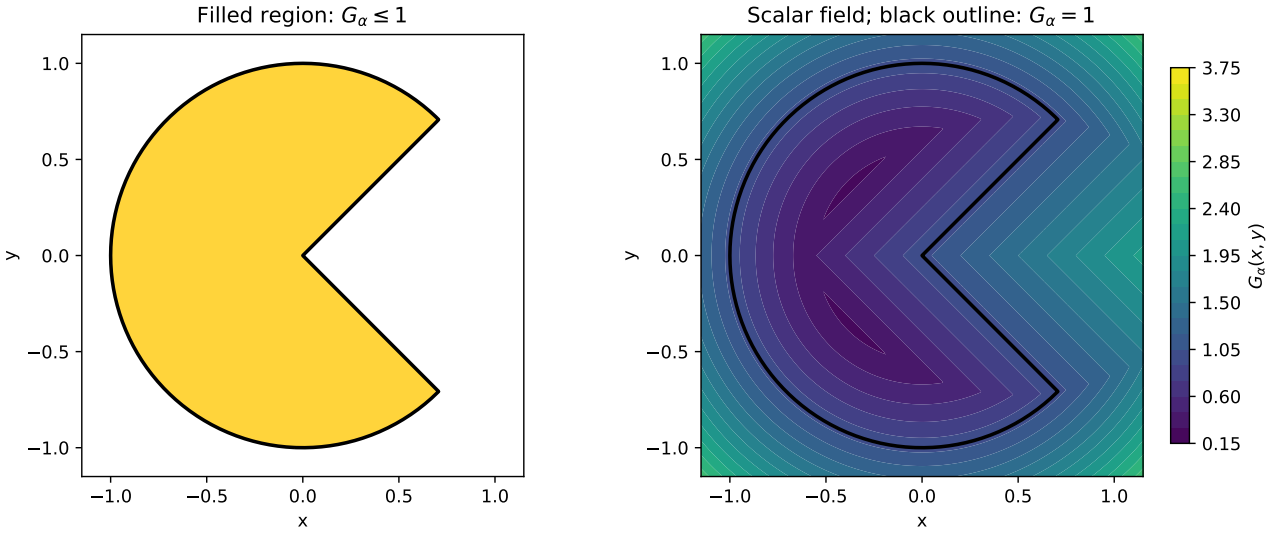


Figure 1: The 90°-mouth example, computed from G itself. Left: $G \leq 1$. Right: field values, with the contour $G = 1$ emphasized.

3 Relation to the previous formula

The earlier note used

$$F_\alpha(x, y) = \max(x^2 + y^2 - 1,\ (\tan\alpha)x - |y|).$$

The identity $1 + \max(A, B) = \max(A + 1, B + 1)$ proves that

$$G_\alpha = F_\alpha + 1, \qquad F_\alpha = 0 \iff G_\alpha = 1, \qquad F_\alpha \leq 0 \iff G_\alpha \leq 1.$$

Thus the new presentation is a compact, exact *level-one implicit equation*. Both versions already use finite elementary expressions with a maximum; the change is a normalization and presentation choice, not a stronger notion of closed form. The old three-dimensional surface moves upward by exactly one unit.

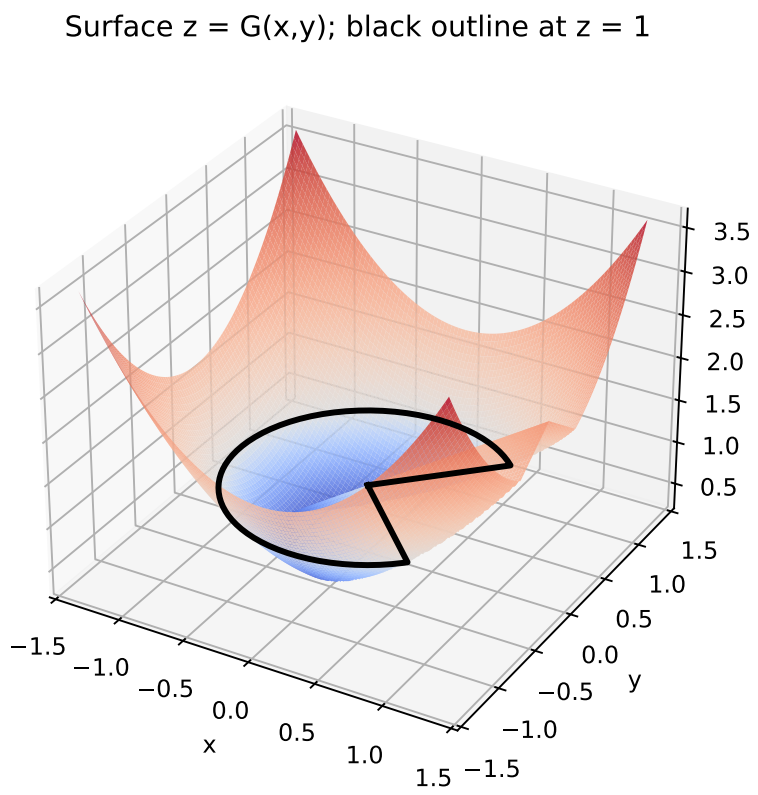


Figure 2: The graph $z = G_{\pi/4}(x, y)$. The black curve lies at height 1. The mouth is raised above that level, while the body interior lies below it.

4 Geometry, regularity, and transformations

For a unit radius, with angles in radians,

$$\text{Area}(P_\alpha) = \pi - \alpha, \qquad \text{Perimeter}(P_\alpha) = 2\pi - 2\alpha + 2.$$

The area subtracts a sector of angle 2α; the perimeter consists of the remaining circular arc plus two unit radial segments. At $\alpha = \pi/4$ these are $3\pi/4$ and $3\pi/2 + 2$.

The field is continuous, locally Lipschitz, and piecewise smooth. It is generally nondifferentiable at active absolute-value folds and where distinct active branches of the maximum meet. It is not a signed distance function: $G - 1$ has the correct inside/outside sign, but its magnitude is not generally distance to the boundary. The assumptions exclude $\alpha = 0$, which creates an interior zero-width seam, and $\alpha = \pi/2$, where tangent is undefined.

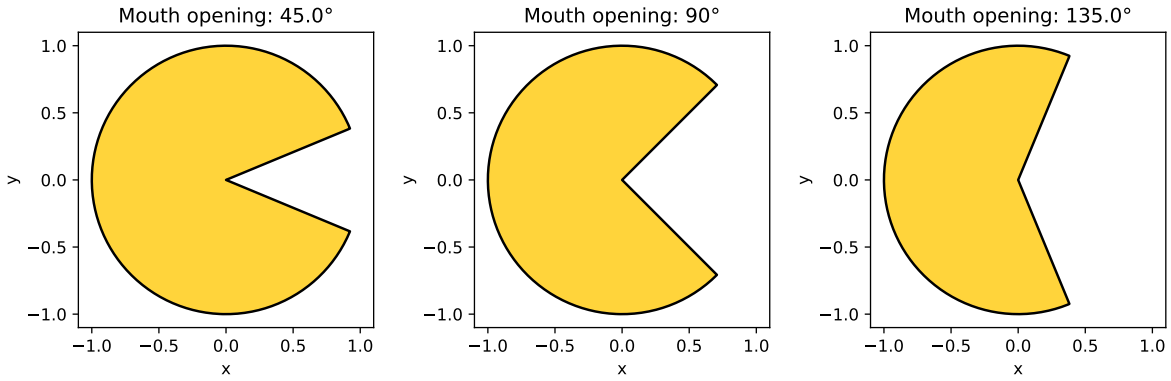


Figure 3: Increasing the half-angle increases the full mouth opening 2α.

For radius $R > 0$, center (c_x, c_y) and mouth direction ϕ, introduce normalized local coordinates

$$u = \frac{\cos\phi(x - c_x) + \sin\phi(y - c_y)}{R}, \qquad v = \frac{-\sin\phi(x - c_x) + \cos\phi(y - c_y)}{R}.$$

Then the transformed outline is still a level-one equation:

$$\boxed{\max(u^2 + v^2,\ 1 + (\tan\alpha)u - |v|) = 1.}$$

Its area is $R^2(\pi - \alpha)$ and perimeter is $R(2\pi - 2\alpha + 2)$. The accompanying scalar evaluator supports these transformations.

5 Code and reproduction

The source bundle contains `pacman.py`, `test_pacman.py`, this LaTeX source, a README, and the generated figures. Python 3.8 or later is supported. The scalar evaluator uses only the standard library. NumPy and Matplotlib are used for plotting. From the extracted source directory, run:

```
python -m pip install -r requirements.txt
python pacman.py --half-angle 45 --output figures
python pacman.py --half-angle 30 --output figures_60deg
python pacman.py --point 0.5 0
python -m unittest -v
pdflatex pacman_note.tex
pdflatex pacman_note.tex
```

The command-line half-angle is in degrees; the Python API takes radians. Plots use the actual maximum expression at level 1, rather than substituting an unrelated silhouette. The 3D overlay uses the analytic boundary for precision. Grid contours are numerical approximations and can require higher resolution for very narrow or wide mouths. The `--resolution` option controls this. Floating-point boundary membership should use a tolerance instead of exact equality.

The tests check arc and mouth-edge samples for five angles, compare 2,000 seeded random points against an independent polar-geometry membership test, and check transformed boundaries and invalid parameters.

6 Relation to existing literature

The level-one formula (1), equivalently the earlier field $F_\alpha = G_\alpha - 1$, sits at the intersection of several established ideas: Pac-Man-shaped domains, level-set descriptions of geometry, oriented or approximate

distance functions, and R-function/constructive-solid-geometry combinations. The literature below is ordered from the closest geometric precedents to broader implicit-function frameworks.

Caboussat, Glowinski, and Gourzoulidis (2022). In numerical experiments for a nonsmooth prescribed Jacobian equation, the authors define the unit disk Ω_d and a "pacman" domain

$$\Omega_p = \Omega_d \setminus \{(x_1, x_2) : x_1 > 0,\ |x_2| < x_1\}.$$

This is the same geometry as $P_{\pi/4}$ in (3), up to open/closed-boundary conventions. Their paper expresses the object as a set difference rather than as the zero/sublevel set of the single scalar function $\max(x^2 + y^2 - 1, x - |y|)$ [1].

Levenberg and Ma'u (2020). Their paper *Monge–Ampère of Pac-Man* studies a nonconvex Pac-Man set described as a disk minus a centrally symmetric wedge. Their principal example has a wedge opening of $\pi/2$ and a mouth vertex at the disk center. After translation of coordinates, this is again the same basic geometry as the $\alpha = \pi/4$ case here; their angle parameter denotes the full wedge opening, whereas α in this note is the half-angle [2].

Beneš (2021). Beneš studies planar "Pacman domains" of the form

$$D_\beta(n) = \{re^{i\theta} : 0 < \theta < 2\pi - \beta,\ 0 < r < 2n\} - z_0,$$

for estimates on discrete Green's functions. This is another disk-with-wedge-removed family, written in polar/complex notation and at a different scale and translation [3].

Jankowski and Stanberry (2012). In work on oriented distance functions for random sets, the authors give an explicit example called "pacman in $\mathbb{R}^2$": a disk centered at the origin with one quadrant removed. They then study the Pac-Man through its *oriented distance function* (ODF), so the shape is encoded by a single scalar field whose zero set is the boundary. The particular mouth orientation and angle differ from (1), and the ODF is not the max expression used here, but this is a direct precedent for treating a two-dimensional Pac-Man as a scalar implicit object rather than only as a drawn set [5].

Mulroy, Lopez, Spenko, and Srivastava (2022). These authors use R-functions and approximate distance fields to control the shape of soft robots. One of their explicit demonstrations is a Pac-Man-like shape. They state that the distance function for that example is constructed by R-equivalence of two line segments and an arc, yielding a single analytical scalar field for the desired outline. This is very close in spirit to the present note: simple boundary primitives are combined into one implicit function, although their smooth approximate-distance construction is different from the nonsmooth max/absolute-value formula (1) [6].

Schaffer and Exl (2024). In a constructive-solid-geometry framework based on R-functions, Schaffer and Exl explicitly discuss Boolean combinations of implicit geometries and include min/max systems among the R-function choices. Their code example constructs a three-dimensional "Pacman" by subtracting a mouth from a sphere, and they plot the zero level set of the resulting approximate distance function. Thus the paper contains the same broad recipe—primitive implicit functions, Boolean-style scalar composition, and a Pac-Man zero level set—though in three dimensions and with a different R-function formula [4].

Neiva and Badia (2021). In their work on h-adaptive aggregated unfitted finite elements, the authors use several embedded interfaces represented as the zero level set of a (piecewise-)smooth scalar function. Their two-dimensional test geometries explicitly include a "pacman" shape, which is also used for a Fichera-corner benchmark. The paper therefore supplies another peer-reviewed instance of

$$\text{Pac-Man boundary} = \{(x, y) : \phi(x, y) = 0\},$$

although it refers elsewhere for the detailed level-set formula and does not print the compact max expression (1) [7].

Burman, Claus, Hansbo, Larson, and Massing (2015). The CutFEM framework represents embedded geometry by level-set functions on a background mesh. As a basic example, the paper uses

$$\phi(x, y) = x^2 + y^2 - 1$$

for a circular domain. This is exactly the first primitive appearing in (1). CutFEM is not itself a Pac-Man construction, but it provides a standard numerical-analysis setting in which a scalar function's sign encodes the inside, outside, and boundary of a planar region [8].

Earlier R-function background. Gladkov and Svendsen present the R-functions method for solving elliptic PDEs on complex two-dimensional domains, illustrating the longer mathematical tradition of building exact or approximate analytical domain descriptions from simpler implicit pieces [9]. This paper is background rather than an exact Pac-Man match, but it helps place the later Pac-Man R-function examples in context.

Smooth polar Pac-Man curves. A non-peer-reviewed Mathematics Stack Exchange discussion asks for smooth polar equations $r = r(\theta)$ that trace a Pac-Man-like outline. Those constructions are conceptually different: they parameterize a curve radially, while (1) represents the filled region and its boundary as sublevel and level-one sets of a scalar field on $\mathbb{R}^2$ [10].

The nonconstant wedge term can itself be rewritten as a min of two half-plane functions,

$$(\tan\alpha)x - |y| = \min((\tan\alpha)x - y,\ (\tan\alpha)x + y)\,.$$

Consequently,

$$F_\alpha(x, y) = \max\bigl(x^2 + y^2 - 1, \min((\tan\alpha)x - y, (\tan\alpha)x + y)\bigr)\,,$$

which makes the constructive-solid-geometry connection especially transparent: a circular implicit primitive is combined with two linear half-plane primitives using min/max operations.

The references establish the geometry and the general scalar-composition method. The earlier note did not locate an exact match for its compact F_α formula in the sources reviewed. The present level-one equation is algebraically equivalent to that formula. No first-publication or novelty claim is made for either form; absence of a located match does not establish priority. The literature discussion and bibliography below are retained from the earlier note, not presented as a new exhaustive search.

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